

Python: exercises on functions

This notebook was generated in **solutions** mode.

Exercise 1: Greeting Function

Exercise Description

Create a function `greet` that takes a user's name as input and prints a greeting message. The function should have a default parameter that uses "Fabio" as the default name if no name is provided.

The function should print: "Hello, [name]! Welcome!"

Your solution

```
def greet(name="Fabio"):
    print("Hello, "+name+"! Welcome!")
```

Test your solution

```
# Test the function with default parameter
import io
import sys
from contextlib import redirect_stdout

# Capture output from greet() with default parameter
f = io.StringIO()
with redirect_stdout(f):
    greet()
output = f.getvalue().strip()
assert output == "Hello, Fabio! Welcome!"
```

Test your solution

```
# Test the function with custom name
f = io.StringIO()
with redirect_stdout(f):
    greet("Alice")
output = f.getvalue().strip()
assert output == "Hello, Alice! Welcome!"
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 2: Even or Odd Checker Function

Exercise Description

Write a function `is_even` that takes an integer as input and returns `True` if the number is even, `False` if it's odd.

Your solution

```
def is_even(number):  
    return number % 2 == 0
```

Test your solution

```
# Test with even number  
assert is_even(4) == True  
assert is_even(0) == True  
assert is_even(100) == True
```

Test your solution

```
# Test with odd number  
assert is_even(3) == False  
assert is_even(1) == False  
assert is_even(99) == False
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 3: Maximum of Two Numbers

Exercise Description

Create a function `max_of_two` that returns the larger of two numbers. The second parameter should have a default value of 0.

Your solution

```
def max_of_two(a, b=0):  
    if a > b:  
        return a  
    else:  
        return b
```

Test your solution

```
# Test with two positive numbers
assert max_of_two(5, 3) == 5
assert max_of_two(2, 8) == 8
assert max_of_two(10, 10) == 10
```

Test your solution

```
# Test with default parameter
assert max_of_two(5) == 5   # 5 > 0 (default)
assert max_of_two(-3) == 0  # -3 < 0 (default)
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 4: Simple Calculator Function

Exercise Description

Implement a function `calculator` that takes two numbers and an operator (+, -, *, /) and returns the result of the operation. Handle division by zero and invalid operators

appropriately:

- If the operator is invalid, return "Invalid operator".
- If division by zero is attempted, return "Division by zero error".

Your solution

```
def calculator(a, b, operator):  
    if operator == '+':  
        return a + b  
    elif operator == '-':  
        return a - b  
    elif operator == '*':  
        return a * b  
    elif operator == '/':  
        if b != 0:  
            return a / b  
        else:  
            return "Error: Division by zero"  
    else:  
        return "Invalid operator"
```

Test your solution

```
# Test basic operations  
assert calculator(10, 5, '+') == 15  
assert calculator(10, 5, '-') == 5  
assert calculator(10, 5, '*') == 50  
assert calculator(10, 5, '/') == 2.0
```

Test your solution

```
# Test error cases
assert calculator(10, 0, '/') == "Error: Division by zero"
assert calculator(10, 5, '%') == "Invalid operator"
```

Debug your solution

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```
from IPython.display import HTML; import urllib.parse as u; HTML('<a href="https://pyth
```

Exercise 5: Temperature Converter Function

Exercise Description

Write a function `celsius_to_fahrenheit` that converts a temperature from Celsius to Fahrenheit using the formula: $F = (C \times 9/5) + 32$

Your solution

```
def celsius_to_fahrenheit(celsius):  
    return (celsius * 9/5) + 32
```

Test your solution

```
# Test common temperature conversions  
assert celsius_to_fahrenheit(0) == 32.0 # Freezing point  
assert celsius_to_fahrenheit(100) == 212.0 # Boiling point  
assert celsius_to_fahrenheit(25) == 77.0 # Room temperature
```

Test your solution

```
# Test negative temperatures  
assert celsius_to_fahrenheit(-10) == 14.0  
assert celsius_to_fahrenheit(-40) == -40.0 # Special case where C = F
```

Debug your solution

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```
from IPython.display import HTML; import urllib.parse as u; HTML('<a href="https://pyth
```

Exercise 6: Factorial Calculator (Recursive)

Exercise Description

Create a recursive function `factorial` that returns the factorial of a given number. The function should have a default parameter with the value 10. Remember that factorial of n is $n! = n \times (n-1) \times (n-2) \times \dots \times 1$, and $0! = 1! = 1$.

Your solution

```
def factorial(n=10):  
    if n == 0 or n == 1:  
        return 1  
    else:  
        return n * factorial(n-1)
```

Test your solution

```
# Test base cases  
assert factorial(0) == 1  
assert factorial(1) == 1
```

Test your solution

```
# Test small factorials  
assert factorial(3) == 6   # 3! = 3 × 2 × 1 = 6  
assert factorial(4) == 24 # 4! = 4 × 3 × 2 × 1 = 24  
assert factorial(5) == 120 # 5! = 5 × 4 × 3 × 2 × 1 = 120
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 7: Leap Year Checker Function

Exercise Description

Write a function `is_leap_year` that determines whether a given year is a leap year. A leap year is divisible by 4, but if it's divisible by 100, it must also be divisible by 400.

Your solution

```
def is_leap_year(year):  
    if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):  
        return True  
    else:  
        return False
```

Test your solution

```
# Test leap years  
assert is_leap_year(2024) == True # Divisible by 4, not by 100
```

```
assert is_leap_year(2000) == True # Divisible by 400
assert is_leap_year(1600) == True # Divisible by 400
```

Test your solution

```
# Test non-leap years
assert is_leap_year(2023) == False # Not divisible by 4
assert is_leap_year(1900) == False # Divisible by 100 but not by 400
assert is_leap_year(2100) == False # Divisible by 100 but not by 400
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 8: Circle Area Calculator

Exercise Description

Implement a function `circle_area` that calculates the area of a circle given its radius. Use the formula: $\text{Area} = \pi \times \text{radius}^2$

Your solution

```
import math

def circle_area(radius):
    return math.pi * radius**2
```

Test your solution

```
# Test with common radius values
import math
assert abs(circle_area(1) - math.pi) < 0.0001 # Area of unit circle
assert abs(circle_area(2) - 4 * math.pi) < 0.0001 # Area = 4π
```

Test your solution

```
# Test with larger radius
assert abs(circle_area(5) - 25 * math.pi) < 0.0001 # Area = 25π
assert circle_area(0) == 0 # Zero radius
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 9: Vowel Checker Function

Exercise Description

Create a function `is_vowel` that checks whether a given character is a vowel (a, e, i, o, u) in either uppercase or lowercase.

Your solution

```
def is_vowel(char):  
    vowels = "aeiouAEIOU"  
    return char in vowels
```

Test your solution

```
# Test with lowercase vowels  
assert is_vowel('a') == True  
assert is_vowel('e') == True  
assert is_vowel('i') == True  
assert is_vowel('o') == True  
assert is_vowel('u') == True
```

Test your solution

```
# Test with uppercase vowels and consonants  
assert is_vowel('A') == True
```

```
assert is_vowel('E') == True
assert is_vowel('b') == False
assert is_vowel('Z') == False
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 10: Simple Interest Calculator

Exercise Description

Write a function `simple_interest` that calculates simple interest given principal amount, rate of interest, and time period. Use the formula: $\text{Simple Interest} = (\text{Principal} \times \text{Rate} \times \text{Time}) / 100$

Your solution

```
def simple_interest(principal, rate, time):
    return (principal * rate * time) / 100
```

Test your solution

```
# Test with common values  
assert simple_interest(1000, 5, 2) == 100.0 # $1000 at 5% for 2 years  
assert simple_interest(5000, 10, 1) == 500.0 # $5000 at 10% for 1 year
```

Test your solution

```
# Test with different scenarios  
assert simple_interest(2000, 7.5, 3) == 450.0 # $2000 at 7.5% for 3 years  
assert simple_interest(1000, 0, 5) == 0.0 # 0% interest rate
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 11: Absolute Value Function

Exercise Description

Create a function `absolute` that returns the absolute value of a number (the non-negative value without regard to its sign).

Your solution

```
def absolute(num):  
    if num >= 0:  
        return num  
    else:  
        return -num
```

Test your solution

```
# Test with positive numbers  
assert absolute(5) == 5  
assert absolute(10.5) == 10.5  
assert absolute(0) == 0
```

Test your solution

```
# Test with negative numbers  
assert absolute(-5) == 5  
assert absolute(-10.5) == 10.5  
assert absolute(-100) == 100
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 12: Maximum of Three Numbers

Exercise Description

Write a function `max_of_three` that returns the largest of three numbers.

Your solution

```
def max_of_three(a, b, c):
    if a >= b and a >= c:
        return a
    elif b >= a and b >= c:
        return b
    else:
        return c
```

Test your solution

```
# Test with different maximum positions
assert max_of_three(10, 5, 8) == 10 # First is maximum
```

```
assert max_of_three(3, 15, 7) == 15 # Second is maximum
assert max_of_three(4, 6, 20) == 20 # Third is maximum
```

Test your solution

```
# Test with equal values
assert max_of_three(5, 5, 5) == 5 # All equal
assert max_of_three(10, 10, 8) == 10 # Two equal maximums
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 13: Power Function

Exercise Description

Write a function `power` that calculates the power of a number (base raised to the exponent).

Your solution

```
def power(base, exponent):  
    return base ** exponent
```

Test your solution

```
# Test with common power calculations  
assert power(2, 3) == 8 # 2^3 = 8  
assert power(5, 2) == 25 # 5^2 = 25  
assert power(10, 0) == 1 # Any number to power 0 is 1
```

Test your solution

```
# Test with edge cases  
assert power(3, 1) == 3 # Any number to power 1 is itself  
assert power(2, 4) == 16 # 2^4 = 16
```

Debug your solution

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```
from IPython.display import HTML; import urllib.parse as u; HTML('<a href="https://pyth
```

Exercise 14: Palindrome Checker Function

Exercise Description

Write a function `is_palindrome` that checks whether a given string reads the same forwards and backwards.

Your solution

```
def is_palindrome(text):  
    text = text.lower()  
    return text == text[::-1]
```

Test your solution

```
# Test with palindromes  
assert is_palindrome("racecar") == True  
assert is_palindrome("level") == True  
assert is_palindrome("a") == True # Single character
```

Test your solution

```
# Test with non-palindromes and case variations  
assert is_palindrome("hello") == False  
assert is_palindrome("Racecar") == True # Case insensitive  
assert is_palindrome("python") == False
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 15: Calculate Body Mass Index (BMI)

Exercise Description

Write a function `calculate_bmi(weight, height)` that returns the Body Mass Index given weight (in kilograms) and height (in meters). The BMI is computed as $\text{weight} / (\text{height}^2)$.

Your solution

```
def calculate_bmi(weight, height):  
    bmi = weight / (height**2)  
    return bmi
```

Test your solution

```
# typical values  
assert round(calculate_bmi(70, 1.75), 2) == 22.86  
assert round(calculate_bmi(60, 1.6), 2) == 23.44
```

Test your solution

```
# edge-like scenarios  
assert round(calculate_bmi(90, 2.0), 2) == 22.5  
assert round(calculate_bmi(45, 1.5), 2) == 20.0
```

Debug your solution

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```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 16: Convert Minutes to Hours and Minutes

Exercise Description

Implement a function `convert_minutes(total_minutes)` that takes a non-negative integer number of minutes and returns a tuple `(hours, minutes)` representing the equivalent hours and remaining minutes.

Your solution

```
def convert_minutes(total_minutes):  
    hours = total_minutes // 60  
    minutes = total_minutes % 60  
    return hours, minutes
```

Test your solution

```
# basic cases  
assert convert_minutes(0) == (0, 0)  
assert convert_minutes(59) == (0, 59)  
assert convert_minutes(60) == (1, 0)
```

Test your solution

```
# additional values  
assert convert_minutes(125) == (2, 5)  
assert convert_minutes(240) == (4, 0)
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 17: Minimum of Two Numbers

Exercise Description

Create a function `min_of_two(a, b)` that returns the smaller of two numbers `a` and `b`.

Your solution

```
def min_of_two(a, b):  
    if a < b:  
        return a  
    else:  
        return b
```

Test your solution

```
# typical comparisons  
assert min_of_two(3, 5) == 3  
assert min_of_two(-1, 2) == -1  
assert min_of_two(7, 7) == 7
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**


```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 18: Determine Grade Based on Score

Exercise Description

Write a function `determine_grade(score)` that returns a letter grade based on a numeric score from 0 to 100 inclusive. Use the following scale: A for 90+, B for 80–89, C for 70–79, D for 60–69, F for below 60.

Your solution

```
def determine_grade(score):  
    if score >= 90:  
        return 'A'  
    elif score >= 80:  
        return 'B'  
    elif score >= 70:  
        return 'C'  
    elif score >= 60:  
        return 'D'  
    else:  
        return 'F'
```

Test your solution

```
# boundary checks
assert determine_grade(100) == 'A'
assert determine_grade(90) == 'A'
assert determine_grade(89) == 'B'
assert determine_grade(80) == 'B'
assert determine_grade(79) == 'C'
assert determine_grade(70) == 'C'
assert determine_grade(69) == 'D'
assert determine_grade(60) == 'D'
assert determine_grade(59) == 'F'
assert determine_grade(0) == 'F'
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

Exercise 19: Convert Fahrenheit to Celsius

Exercise Description

Implement a function `fahrenheit_to_celsius(fahrenheit)` that converts a temperature in Fahrenheit to Celsius using the formula $C = (F - 32) * 5 / 9$. Return the Celsius value as

a float.

Your solution

```
def fahrenheit_to_celsius(fahrenheit):  
    return (fahrenheit - 32) * 5 / 9
```

Test your solution

```
# known reference points  
assert round(fahrenheit_to_celsius(32), 2) == 0.00  
assert round(fahrenheit_to_celsius(212), 2) == 100.00  
assert round(fahrenheit_to_celsius(68), 2) == 20.00
```

Test your solution

```
# additional checks  
assert round(fahrenheit_to_celsius(14), 2) == -10.00  
assert round(fahrenheit_to_celsius(41), 2) == 5.00
```

Debug your solution

The following cells contain code to generate links to Python Tutor for debugging your code when running the tests. **You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.**

```
from IPython.display import HTML;import urllib.parse as u;HTML('<a href="https://pyth
```

